1. The charge on a body is 8×10^{-12} C. It means that the body has :

(a) lost 8×10^{-12} electrons

(b) gained 4×10^{10} electrons

(c) gained 2×10^{8} electrons

(d) lost 5×10^{7} electrons

31. (a) (i) What is an electric dipole ? Derive an expression for the torque acting on an electric dipole in a uniform electric field.

(ii) An electric dipole with dipole moment 6×10^{-9} C-m is aligned at an angle of $30°$ with the direction of a uniform electric field of magnitude 4×10^{4} NC^{-1}. Calculate magnitude of the torque acting on the electric dipole. *5*

(b) (i) State Gauss's law in electrostatics. Using it, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius R at a point (i) outside, and (ii) inside the shell.

(ii) A point charge of 4 μC is at the centre of a cubic Gaussian surface, 1·0 m on edge. Find the electric flux through one of the faces of the Gaussian surface. *5*

A small sphere S_1 with charge $-8q$ is $1{\cdot}6$ m away from another identical sphere S_2 with charge $+2q$. The two spheres are brought in contact with each other and then separated by a distance $1{\cdot}6$ m. Initially the force between the two spheres was $8{\cdot}1 \times 10^{-4}$ N.

Based on the above facts, answer the following questions :

(i) Which sphere will transfer the electrons to the other sphere after they were brought in contact ? *1*

(ii) How does the net electric field at the midpoint on the line joining the two spheres change after contact ? *1*

(iii) What was initial charge on spheres S_1 and S_2 ? *2*

OR

(iii) What is the charge on spheres S_1 and S_2 after contact ? *2*

4. A particle of mass m and charge –q is moving with a uniform speed v in a circle of radius r, with another charge q at the centre of the circle. The value of r is :

(a) $\dfrac{1}{4\pi \varepsilon_0 m}\left(\dfrac{q}{v}\right)$

(b) $\dfrac{1}{4\pi \varepsilon_0 m}\left(\dfrac{q}{v}\right)^2$

(c) $\dfrac{m}{4\pi \varepsilon_0}\left(\dfrac{q}{v}\right)$

(d) $\dfrac{m}{4\pi \varepsilon_0}\left(\dfrac{q}{v}\right)^2$

6. A point charge q is kept at a distance r from an infinitely long straight wire with charge density λ. The magnitude of the electrostatic force experienced by charge q is :

(a) Zero

(b) $\dfrac{q\lambda}{2\pi \varepsilon_0 r}$

(c) $\dfrac{q\lambda}{4\pi \varepsilon_0 r}$

(d) $\dfrac{q\lambda}{\varepsilon_0 r}$

1. Two charges q_1 and q_2 are placed at the centres of two spherical conducting shells of radius r_1 and r_2 respectively. The shells are arranged such that their centres are d $[> (r_1 + r_2)]$ distance apart. The force on q_2 due to q_1 is :

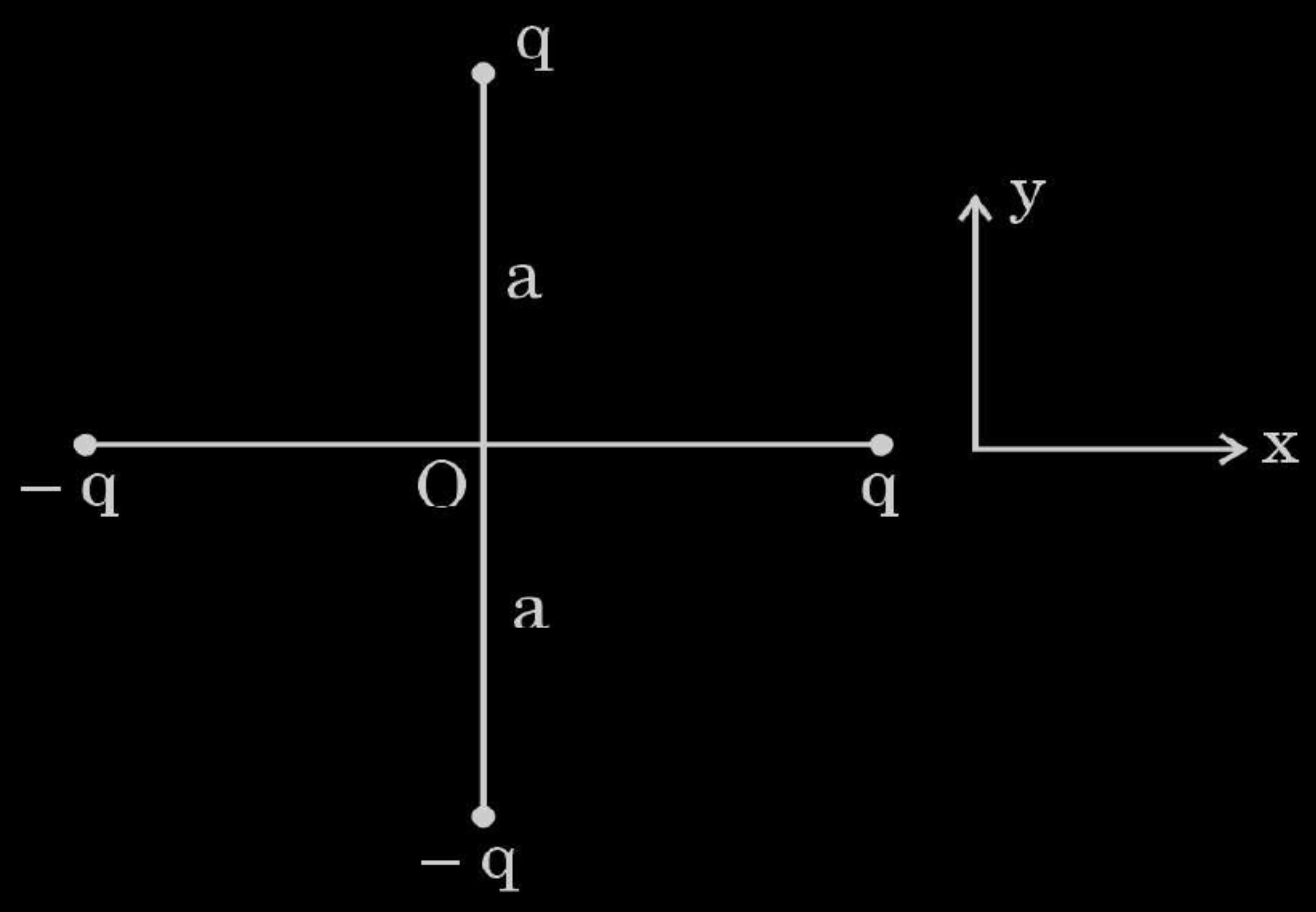
(a) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1q_2}{d^2}$

(b) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1q_2}{(d-r_1)^2}$

(c) Zero

(d) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1q_2}{[d-(r_1+r_2)]^2}$

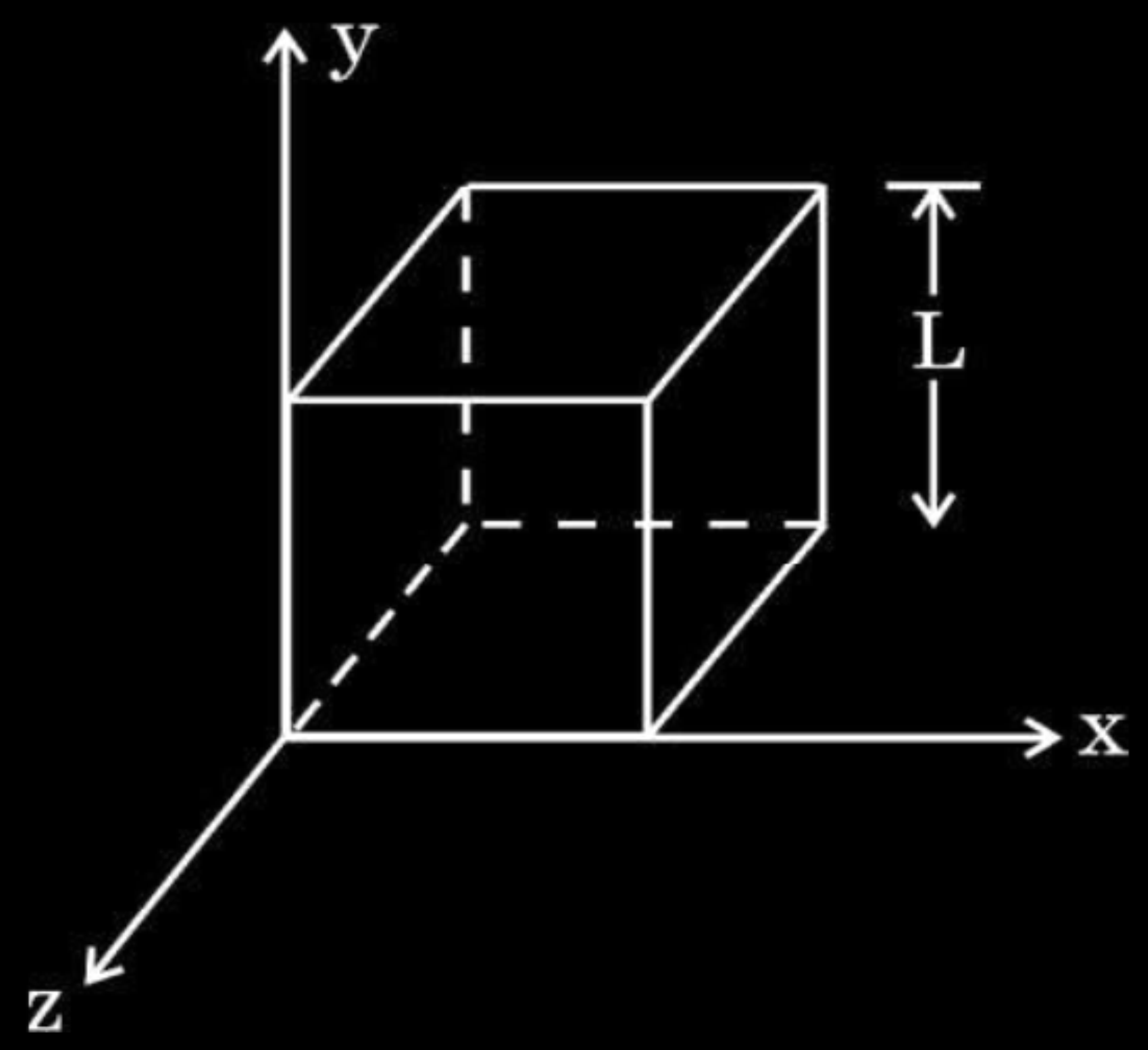
19. Two identical dipoles are arranged in x-y plane as shown in the figure. Find the magnitude and the direction of net electric field at the origin O. *2*



32. (a) (i) Define electric flux and write its SI unit.

(ii) Use Gauss' law to obtain the expression for the electric field due to a uniformly charged infinite plane sheet.

(iii) A cube of side L is kept in space, as shown in the figure. An electric field $\vec{E} = (Ax + B)\,\hat{i}\,\frac{N}{C}$ exists in the region. Find the net charge enclosed by the cube. 5



4. An infinitely long uniformly charged wire produces an electric field of 18×10^4 NC^{-1} at a distance of 1·0 cm. The linear charge density on the wire is :

(a) $1{\cdot}12 \times 10^{-14}$ Cm^{-1}

(b) $3{\cdot}08 \times 10^{-15}$ Cm^{-1}

(c) $1{\cdot}0 \times 10^{-9}$ Cm^{-1}

(d) $1{\cdot}0 \times 10^{-7}$ Cm^{-1}

1. An electron experiences a force $(1\cdot6 \times 10^{-16}\ \mathrm{N})\ \hat{i}$ in an electric field $\vec{E}$. The electric field $\vec{E}$ is :

(a) $(1\cdot0 \times 10^{3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{i}$

(b) $-(1\cdot0 \times 10^{3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{i}$

(c) $(1\cdot0 \times 10^{-3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{i}$

(d) $-(1\cdot0 \times 10^{-3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{i}$

9. A point charge situated at a distance 'r' from a short electric dipole on its axis, experiences a force $\vec{F}$. If the distance of the charge is '2r', the force on the charge will be :

(a) $\frac{\vec{F}}{16}$

(b) $\frac{\vec{F}}{8}$

(c) $\frac{\vec{F}}{4}$

(d) $\frac{\vec{F}}{2}$

32. (a) (i) State Coulomb's law in electrostatics and write it in vector form, for two charges.

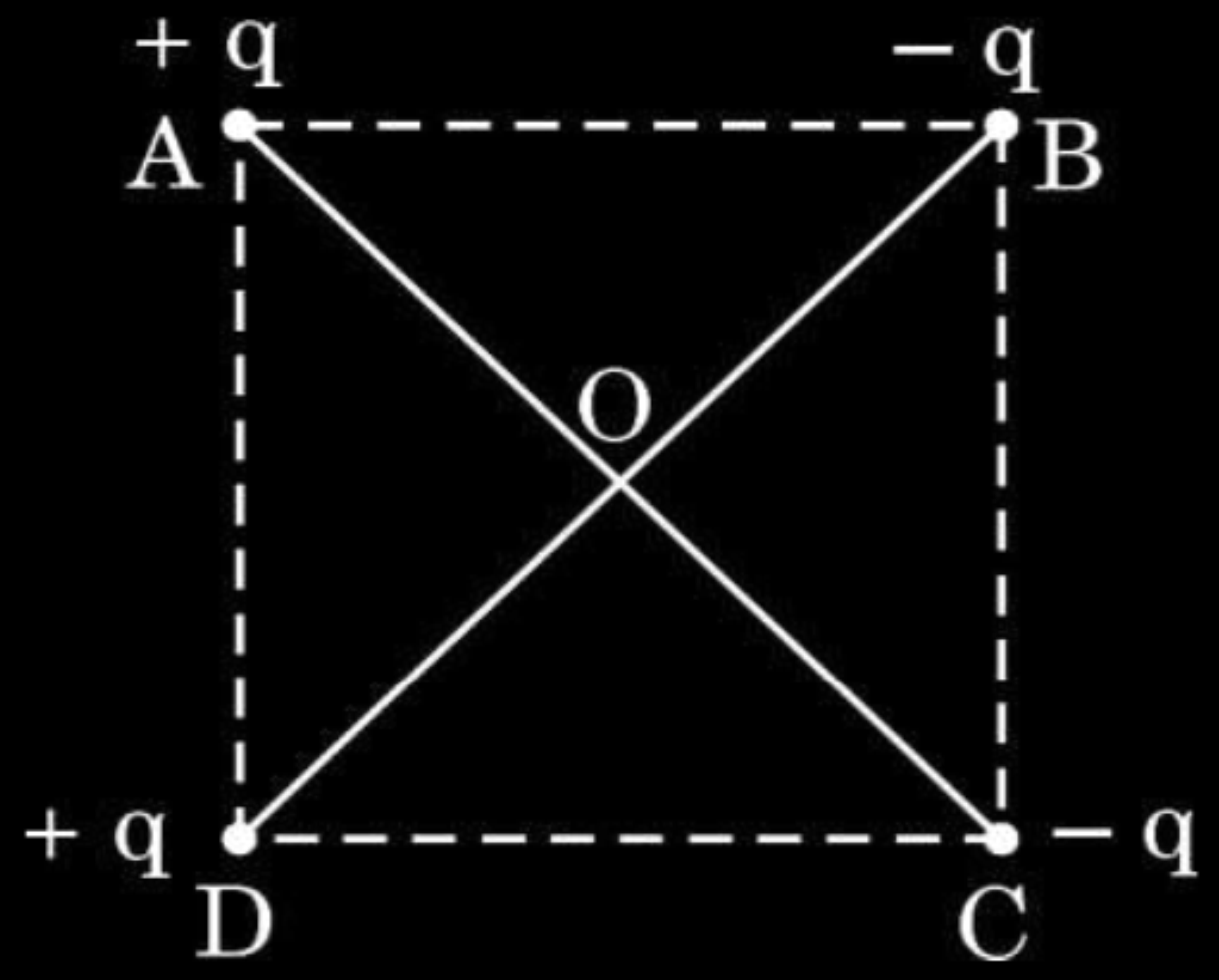
(ii) 'Gauss's law is based on the inverse-square dependence on distance contained in the Coulomb's law.' Explain.

(iii) Two charges A (charge q) and B (charge 2q) are located at points (0, 0) and (a, a) respectively. Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x-axis and y-axis respectively. Find the force exerted by A on B, in terms of $\hat{i}$ and $\hat{j}$. *5*

(b) (i) Derive an expression for the electric field at a point on the equatorial plane of an electric dipole consisting of charges q and – q separated by a distance 2a.

(ii) The distance of a far off point on the equatorial plane of an electric dipole is halved. How will the electric field be affected for the dipole ?

(iii) Two identical electric dipoles are placed along the diagonals of a square ABCD of side $\sqrt{2}$ m as shown in the figure. Obtain the magnitude and direction of the net electric field at the centre (O) of the square. 5



33. (a) (i) Use Gauss' law to obtain an expression for the electric field due to an infinitely long thin straight wire with uniform linear charge density λ.

(ii) An infinitely long positively charged straight wire has a linear charge density λ. An electron is revolving in a circle with a constant speed v such that the wire passes through the centre, and is perpendicular to the plane, of the circle. Find the kinetic energy of the electron in terms of magnitudes of its charge and linear charge density λ on the wire.

(iii) Draw a graph of kinetic energy as a function of linear charge density λ. **5**

9. A point charge q_0 is moving along a circular path of radius a, with a point charge $-Q$ at the centre of the circle. The kinetic energy of q_0 is **1**

(a) $\frac{q_0 Q}{4\pi \in_0 a}$

(b) $\frac{q_0 Q}{8\pi \in_0 a}$

(c) $\frac{q_0 Q}{4\pi \in_0 a^2}$

(d) $\frac{q_0 Q}{8\pi \in_0 a^2}$

7. An electric dipole of dipole moment 2×10^{-8} C-m in a uniform electric field experiences a maximum torque of 6×10^{-4} N-m. The magnitude of electric field is **1**

(a) $2.2 \times 10^{3}\ Vm^{-1}$
(b) $1.2 \times 10^{4}\ Vm^{-1}$
(c) $3.0 \times 10^{4}\ Vm^{-1}$
(d) $4.2 \times 10^{3}\ Vm^{-1}$

5. An isolated point charge particle produces an electric field $\vec{E}$ at a point 3 m away from it. The distance of the point at which the field is $\frac{\vec{E}}{4}$ will be **1**

(a) 2 m (b) 3 m

(c) 4 m (d) 6 m

1. The magnitude of the electric field due to a point charge object at a distance of 4.0 m is 9 N/C. From the same charged object the electric field of magnitude, $16 \frac{N}{C}$ will be at a distance of **1**

(a) 1 m (b) 2 m

(c) 3 m (d) 6 m

1. An electric dipole of length 2 cm is placed at an angle of $30°$ with an electric field 2×10^5 N/C. If the dipole experiences a torque of 8×10^{-3} Nm, the magnitude of either charge of the dipole, is **1**

(A) 4 μC

(B) 7 μC

(C) 8 mC

(D) 2 mC

1. A charge Q is placed at the centre of a cube. The electric flux through one if its face is **1**

(A) $\frac{Q}{\varepsilon_0}$ (B) $\frac{Q}{6\varepsilon_0}$

(C) $\frac{Q}{8\varepsilon_0}$ (D) $\frac{Q}{3\varepsilon_0}$

22. An electric dipole of dipole moment $(\vec{p})$ is kept in a uniform electric field $\vec{E}$. Show graphically the variation of torque acting on the dipole (τ) with its orientation (θ) in the field. Find the orientation in which torque is (i) zero and (ii) maximum. **2**

31. (a) (i) Why do we use a small test charge to measure an electric field ?

(ii) A small stationary positively charged particle is free to move in an electric field. In which direction will it begin to move ?

(iii) Two point charges Q_1 (40 μC) and Q_2 (–16 μC) are placed along x-axis at 0 cm and 24 cm from the origin respectively. Calculate the net force on a third charge Q_3 (– 2·5 μC) placed at x = 36 cm from the origin. *5*

2. 10^9 electrons are transferred to a pith ball with charge $0 \cdot 16$ nC. Its charge now is :

(a) Zero

(b) $-3 \cdot 2 \times 10^{-10}$ C

(c) $-1 \cdot 6 \times 10^{-9}$ C

(d) $3 \cdot 2 \times 10^{-10}$ C

1. The electric flux through a Gaussian spherical surface enclosing a point charge q is ϕ. If the charge is replaced by an electric dipole, magnitude of its dipole moment being 2qa, the flux through the surface will be :

(a) 2ϕ (b) ϕ

(c) $\frac{\phi}{2}$ (d) Zero

22. A uniform electric field is represented as $\vec{E} = (3 \times 10^3 \frac{N}{C})\ \hat{i}$. Find the electric flux of this field through a square of side 10 cm when the :

(a) plane of the square is parallel to y-z plane, and

(b) the normal to plane of the square makes an angle of 60° with the x-axis. *2*

1. An electric dipole with dipole moment $\vec{P} = P_0 \hat{i} - P_0 \hat{j}$ is placed in an electric field $\vec{E} = E_1 \hat{i} + E_2 \hat{j}$, where P_0, E_1 and E_2 are constants. The torque $\vec{\tau}$ acting on the dipole is :

(a) $P_0 (E_2 - E_1) \hat{k}$

(b) $P_0 (E_2 + E_1) \hat{k}$

(c) $-P_0 (E_2 + E_1) \hat{k}$

(d) $P_0 (E_1 - E_2) \hat{k}$

22. The inward and the outward electric flux through a Gaussian surface are 2ϕ and ϕ respectively.

(a) What is the net charge enclosed by the surface ?

(b) If the net outward flux through the surface were zero, can it be concluded that there were no charges inside the surface ? Justify your answer. *2*

18. (a) Four point charges of 1 μC, –2 μC, 1 μC and –2 μC are placed at the corners A, B, C and D respectively, of a square of side 30 cm. Find the net force acting on a charge of 4 μC placed at the centre of the square.

18. (b) Three point charges, 1 pC each, are kept at the vertices of an equilateral triangle of side 10 cm. Find the net electric field at the centroid of triangle.

33. (b) (i) A thin spherical shell of radius R has a uniform surface charge density σ. Using Gauss' law, deduce an expression for electric field (i) outside and (ii) inside the shell.

(ii) Two long straight thin wires AB and CD have linear charge densities 10 μC/m and -20 μC/m, respectively. They are kept parallel to each other at a distance 1 m. Find magnitude and direction of the net electric field at a point midway between them.

23. (a) Define the term 'electric flux' and write its dimensions.

(b) A plane surface, in shape of a square of side 1 cm is placed in an electric field $\vec{E} = \left(100 \frac{N}{C}\right)\hat{i}$ such that the unit vector normal to the surface is given by $\hat{n} = 0.8\hat{i} + 0.6\hat{k}$. Find the electric flux through the surface.

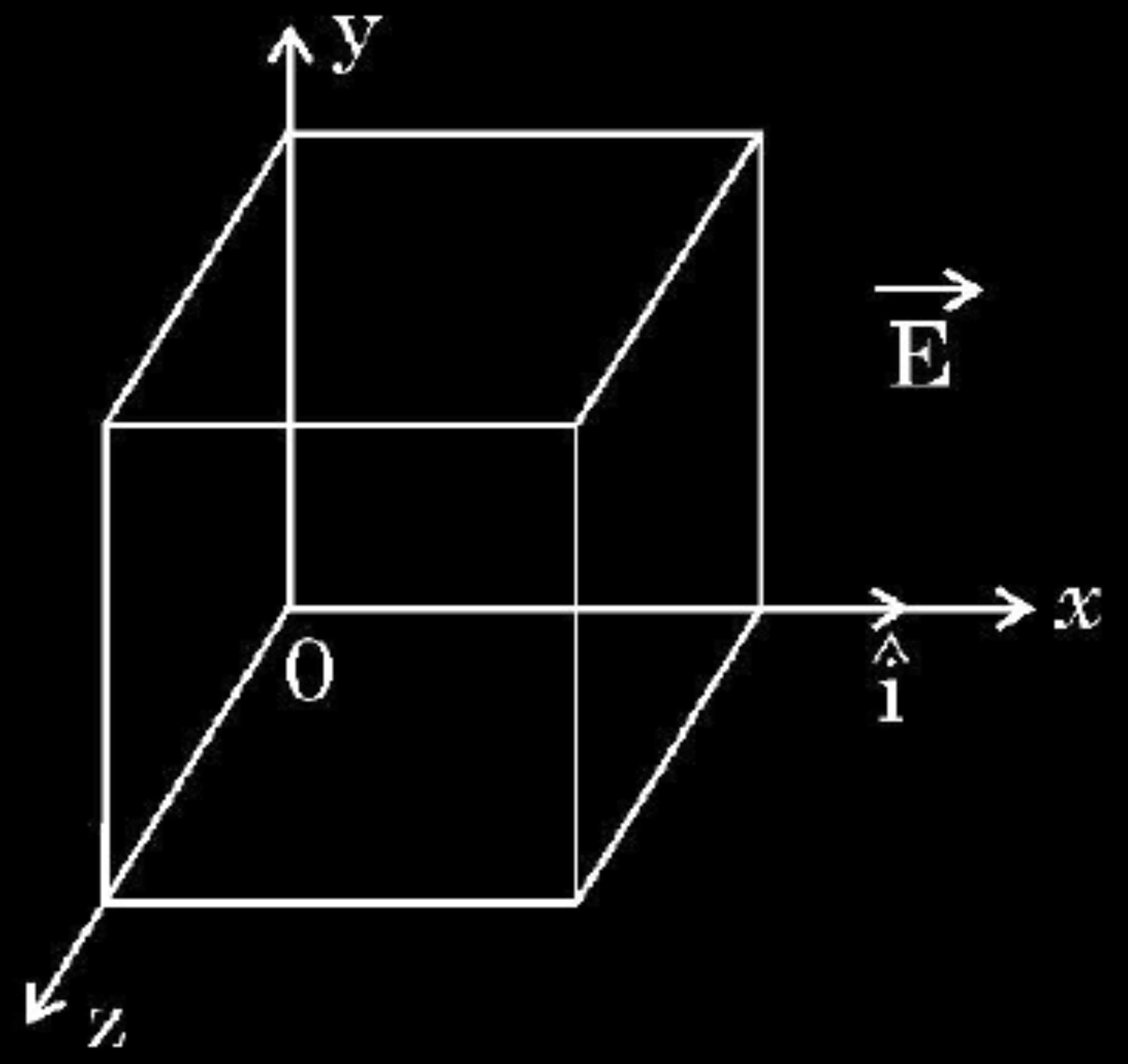
(b) (i) Using Gauss's law, show that the electric field $\vec{E}$ at a point due to a uniformly charged infinite plane sheet is given by $\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$ where symbols have their usual meanings.

(ii) Electric field $\vec{E}$ in a region is given by

$$\vec{E} = (5x^2 + 2)\,\hat{i}$$

where E is in N/C and x is in meters.

A cube of side 10 cm is placed in the region as shown in figure.



Calculate (1) the electric flux through the cube, and (2) the net charge enclosed by the cube. **5**

2. An infinite long straight wire having a charge density λ is kept along $y'y$ axis in x-y plane. The Coulomb force on a point charge q at a point P $(x, 0)$ will be **1**

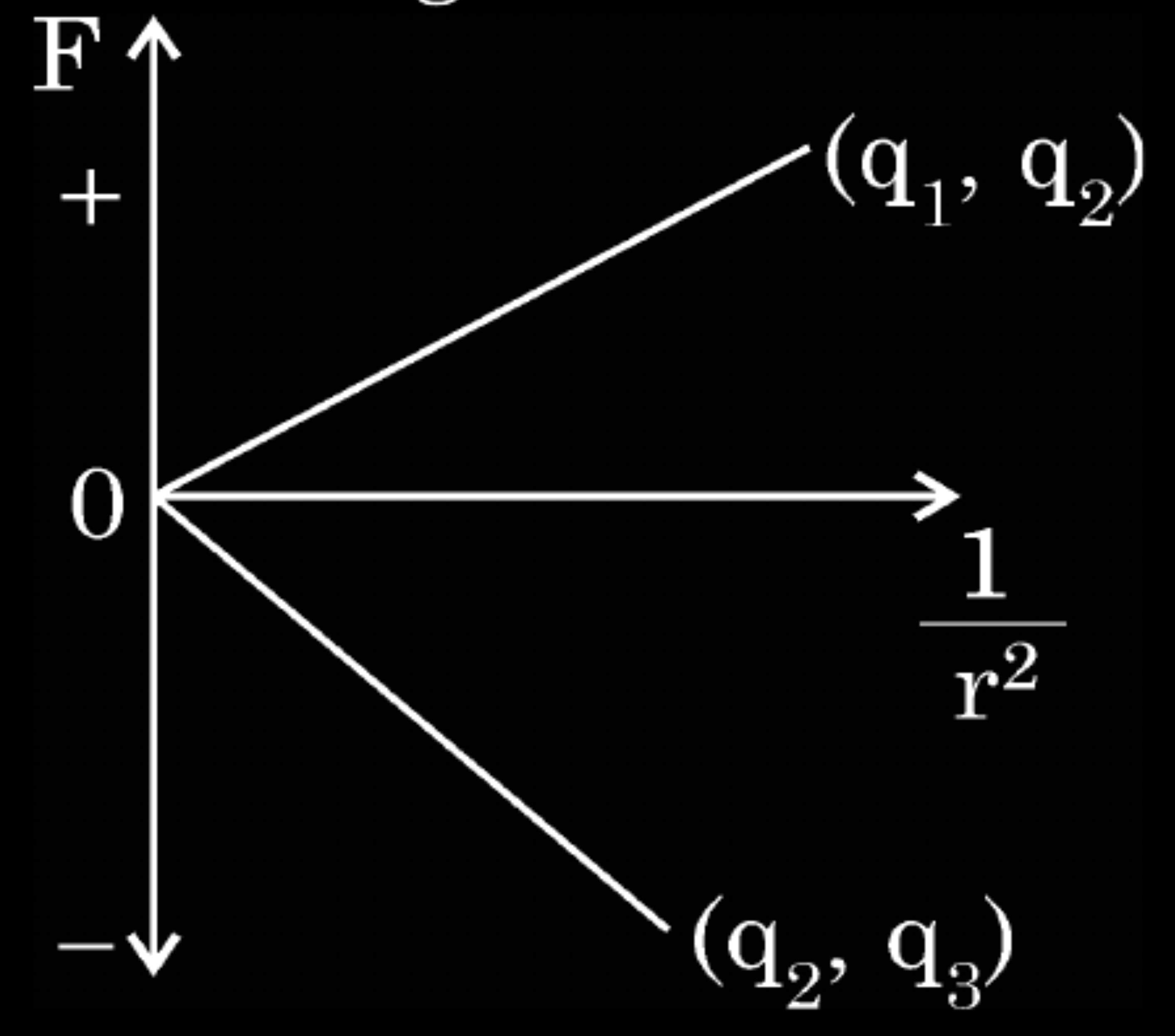
(A) attractive and $\frac{q\lambda}{2\pi\varepsilon_0 x}$

(B) repulsive and $\frac{q\lambda}{2\pi\varepsilon_0 x}$

(C) attractive and $\frac{q\lambda}{\pi\varepsilon_0 x}$

(D) repulsive and $\frac{q\lambda}{\pi\varepsilon_0 x}$

2. The Coulomb force (F) versus $(1/r^2)$ graphs for two pairs of point charges (q_1 and q_2) and (q_2 and q_3) are shown in figure. The charge q_2 is positive and has least magnitude. Then **1**



(A) $q_1 > q_2 > q_3$

(B) $q_1 > q_3 > q_2$

(C) $q_3 > q_2 > q_1$

(D) $q_3 > q_1 > q_2$

2. Two identical small conducting balls B_1 and B_2 are given –7 pC and + 4 pC charges respectively. They are brought in contact with a third identical ball B_3 and then separated. If the final charge on each ball is –2 pC, the initial charge on B_3 was **1**

(A) –2 pC (B) –3 pC

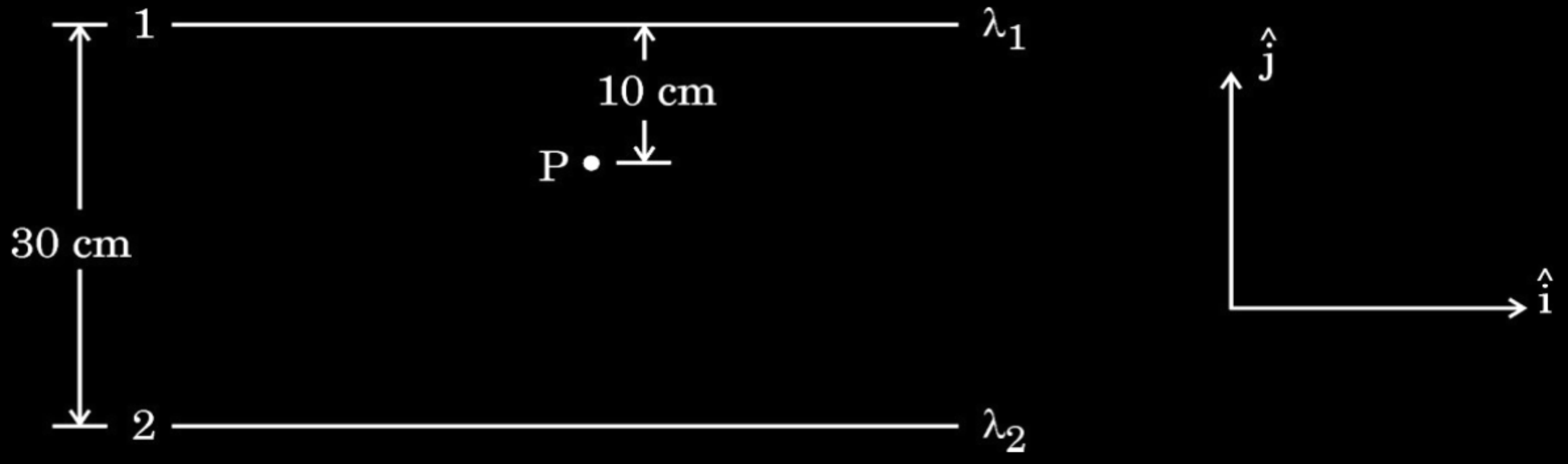
(C) –5 pC (D) –15 pC

(b) (i) A charge + Q is placed on a thin conducting spherical shell of radius R. Use Gauss's theorem to derive an expression for the electric field at a point lying (i) inside and (ii) outside the shell.

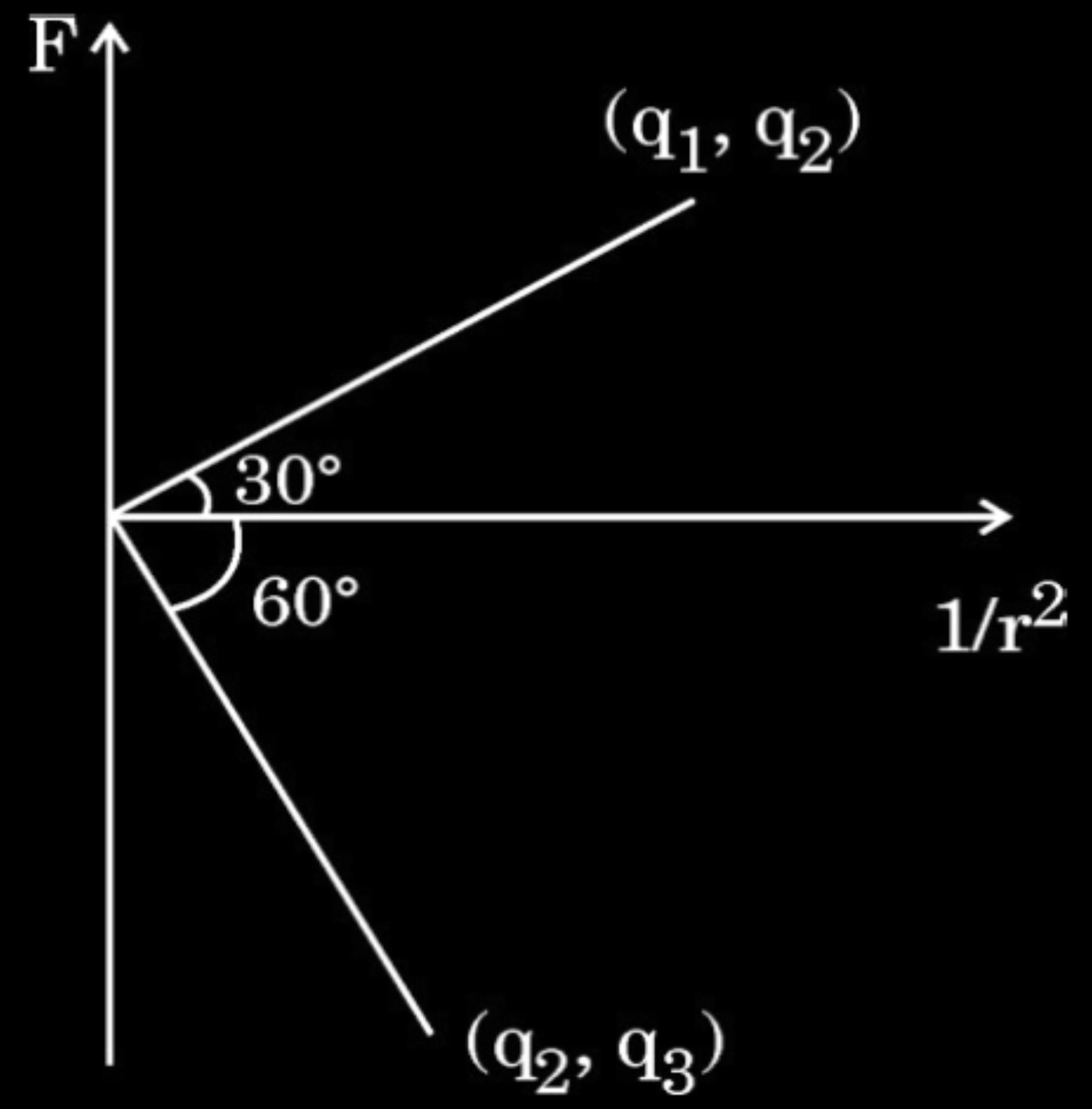
(ii) Show that the electric field for same charge density (σ) is twice in case of a conducting plate or surface than in a nonconducting sheet. *5*

(b) (i) State Gauss's Law in electrostatics. Apply this to obtain the electric field $\vec{E}$ at a point near a uniformly charged infinite plane sheet.

(ii) Two long straight wires 1 and 2 are kept as shown in the figure. The linear charge density of the two wires are $\lambda_1 = 10\ \mu C/m$ and $\lambda_2 = -20\ \mu C/m$. Find the net force $\vec{F}$ experienced by an electron held at point P. 5



1. Coulomb force F versus $\left(\frac{1}{r^2}\right)$ graphs for two pairs of point charges (q_1 and q_2) and (q_2 and q_3) are shown in the figure. The ratio of charges $\left(\frac{q_1}{q_3}\right)$ is :



(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) 3

(D) $\frac{1}{3}$

1. Two charged particles P and Q, having the same charge but different masses m_P and m_Q, start from rest and travel equal distances in a uniform electric field $\vec{E}$ in time t_P and t_Q respectively. Neglecting the effect of gravity, the ratio $\left(\frac{t_P}{t_Q}\right)$ is :

(A) $\frac{m_P}{m_Q}$

(B) $\frac{m_Q}{m_P}$

(C) $\sqrt{\frac{m_P}{m_Q}}$

(D) $\sqrt{\frac{m_Q}{m_P}}$

(ii) An electric dipole consists of point charges $-1\cdot0$ pC and $+1\cdot0$ pC located at (0, 0) and (3 mm, 4 mm) respectively in $x-y$ plane. An electric field $\vec{E} = \left(\frac{1000\ \text{V}}{\text{m}}\right)\hat{i}$ is switched on in the region. Find the torque $\vec{\tau}$ acting on the dipole.

1. A thin plastic rod is bent into a circular ring of radius R. It is uniformly charged with charge density λ. The magnitude of the electric field at its centre is :

(A) $\frac{\lambda}{2\varepsilon_0 R}$ (B) Zero (C) $\frac{\lambda}{4\pi\varepsilon_0 R}$ (D) $\frac{\lambda}{4\varepsilon_0 R}$

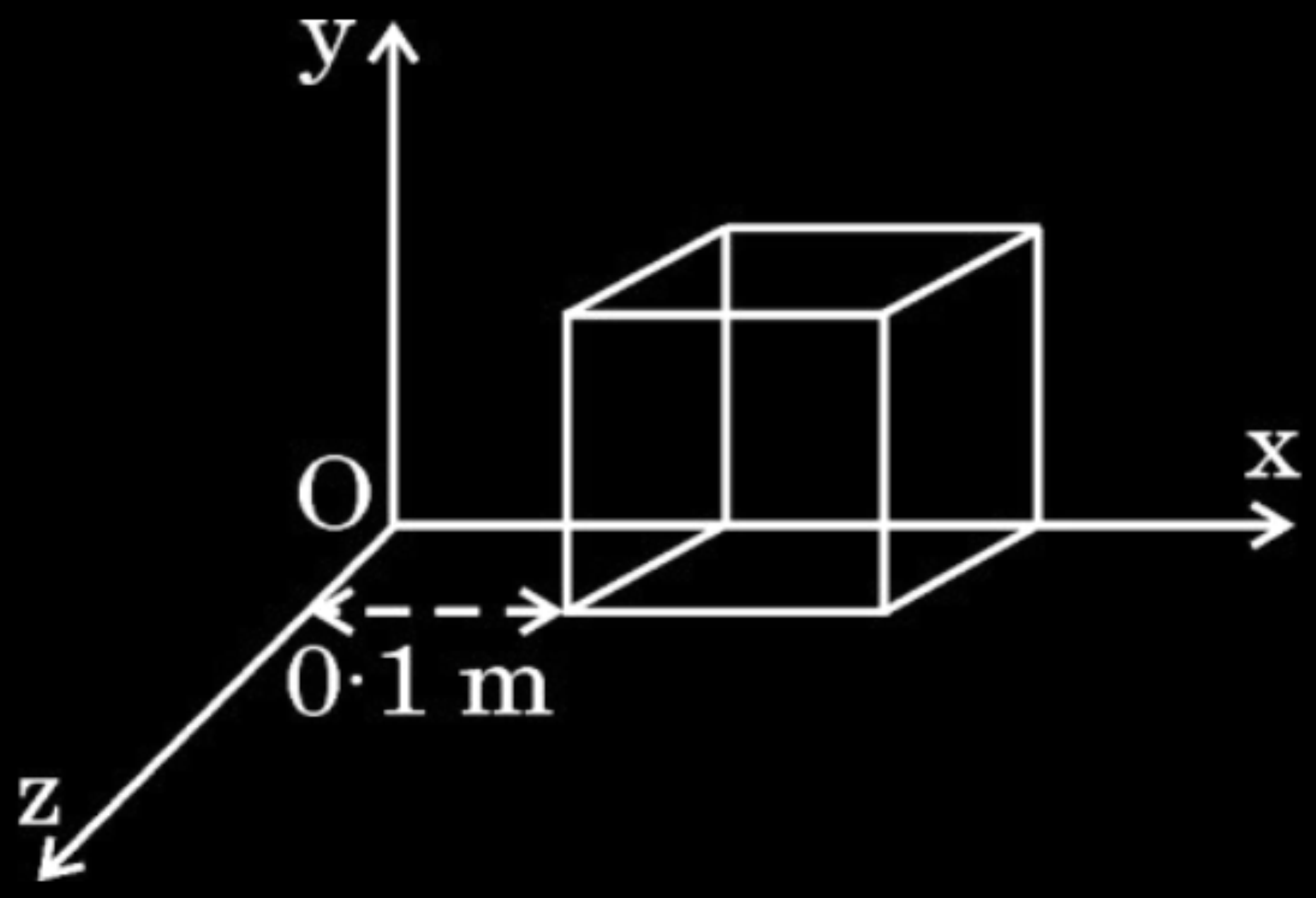
2. A charged sphere of radius r has surface charge density σ. The electric field on its surface is E. If the radius of the sphere is doubled, keeping charge density the same, the ratio of the electric field on the old sphere to that on the new sphere will be :

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) 4

22. A cube of side 0·1 m is placed, as shown in the figure, in a region where electric field $\vec{E} = 500\, x\, \hat{i}$ exists. Here x is in meters and E in NC^{-1}. Calculate : *3*

(a) the flux passing through the cube, and

(b) the charge within the cube.



30. An electric dipole is a pair of equal and opposite point charges + q and – q separated by a distance 2a. A dipole can be characterised by dipole moment determined by its charge and separation between the two charges. It is a vector quantity. By convention the direction of a dipole is taken to be the direction from – q to + q. Like fields due to a point charge, fields due to a given dipole at a point can be found.

When a dipole is placed in a uniform external electric field, it experiences a torque. This torque tends to rotate the dipole and work is done in this process. This work is stored as the potential energy of the system.

(i) An electric dipole is kept in a uniform electric field such that the dipole moment is not along the electric field. It will experience : *1*

(A) A net force and a torque

(B) A net force but no torque

(C) A torque but no net force

(D) neither a net force nor a torque

(ii) An electric dipole consists of charges + q and – q, separated by distance 2a. The force acting on a point charge placed at a distance x (>> a) on the axis of the dipole is F. If the distance of the charge is doubled, the force will become : *1*

(A) $\frac{F}{16}$ (B) $\frac{F}{8}$

(C) $\frac{F}{4}$ (D) $\frac{F}{2}$

(iii) Two identical small electric dipoles, each of dipole moment $\vec{p}$ are arranged perpendicular to each other such that their negative charges coincide. The magnitude of the net dipole moment of the arrangement will be : *1*

(A) p (B) 2p

(C) $p\sqrt{2}$ (D) $\frac{p}{\sqrt{2}}$

(iv) (a) A molecule with dipole moment $3 \cdot 0 \times 10^{-30}$ C-m is placed in a uniform electric field of 2×10^{4} NC^{-1}. The maximum torque experienced by the molecule is : *1*

(A) $1 \cdot 5 \times 10^{-34}$ N-m

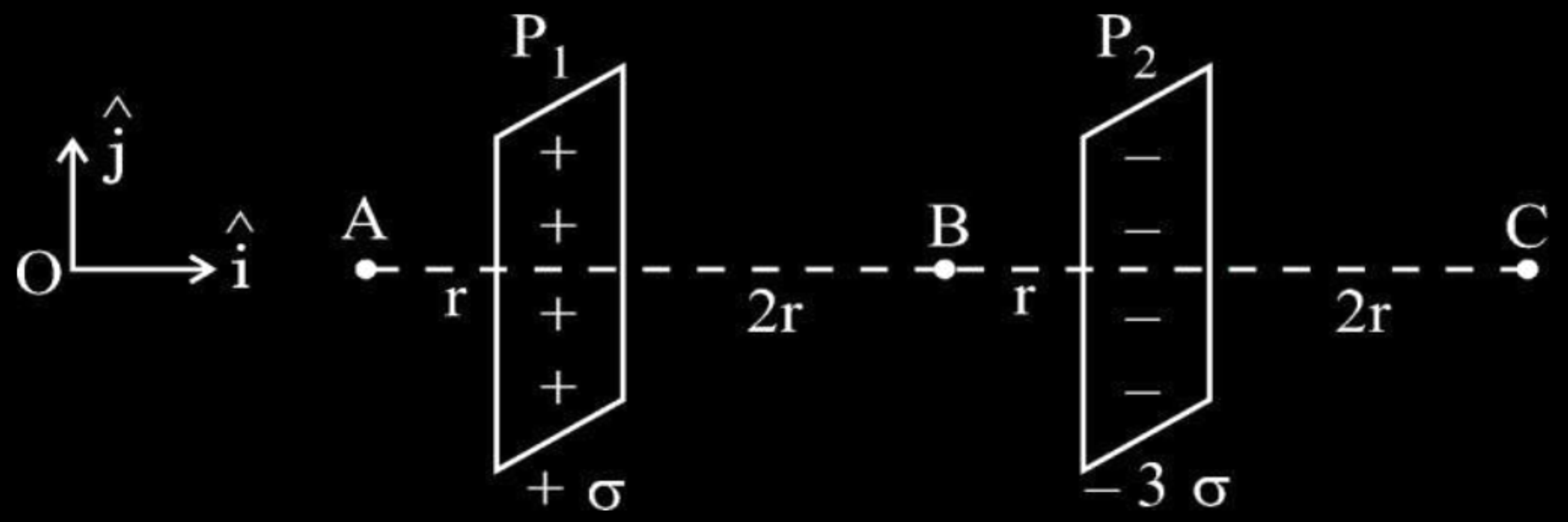
(B) $6 \cdot 0 \times 10^{-26}$ N-m

(C) $4 \cdot 5 \times 10^{34}$ N-m

(D) $6 \cdot 0 \times 10^{-34}$ N-m

(b) Two point charges of 3 μC and 4 μC are kept in air at (0·3 m, 0) and (0, 0.3 m) in x-y plane. Find the magnitude and direction of the net electric field produced at the origin (0, 0). *2*

(b) Two large plane sheets P_1 and P_2 having charge densities $+\sigma$ and -3σ respectively are arranged parallel to each other as shown in the figure. Find the net electric field ($\vec{E}$) at points A, B and C. *3*



(i) Two identical electric dipoles, each consisting of charges $-q$ and $+q$ separated by distance d, are arranged in x-y plane such that their negative charges lie at the origin O and positive charges lie at points (d, 0) and (0, d) respectively. The net dipole moment of the system is : *1*

(A) $-q\,d\,(\hat{i}+\hat{j})$

(B) $q\,d\,(\hat{i}+\hat{j})$

(C) $q\,d\,(\hat{i}-\hat{j})$

(D) $q\,d\,(\hat{j}-\hat{i})$

(ii) E_1 and E_2 are magnitudes of electric field due to a dipole, consisting of charges $-q$ and $+q$ separated by distance $2a$, at points r ($>> a$) (1) on its axis, and (2) on equatorial plane, respectively. Then $\left(\frac{E_1}{E_2}\right)$ is : *1*

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

(iii) An electric dipole of dipole moment 5.0×10^{-8} Cm is placed in a region where an electric field of magnitude 1.0×10^{3} N/C acts at a given instant. At that instant the electric field $\vec{E}$ is inclined at an angle of 30° to dipole moment $\vec{P}$. The magnitude of torque acting on the dipole, at that instant is : *1*

(A) 2.5×10^{-5} Nm

(B) 5.0×10^{-5} Nm

(C) 1.0×10^{-4} Nm

(D) 2.0×10^{-6} Nm

(a) (i) Define electric field. Write its SI units. Derive an expression for electric field due to a point charge at a point. Why must the test charge be as small as possible ?

(ii) An electron and an α-particle, both are kept in the same electric field $\vec{E} = E_1 \hat{i} + E_2 \hat{j}$. Find the force acting on them and their acceleration. *5*

(ii) An electric dipole of dipole moment $\vec{p} = (0{\cdot}8\,\hat{i} + 0{\cdot}6\,\hat{j})\,10^{-29}$ Cm is placed in an electric field $\vec{E} = 1{\cdot}0 \times 10^{7}\,\hat{k}\,\frac{V}{m}$. Calculate the magnitude of the torque acting on it and the angle it makes with the x-axis, at this instant.

23. An electric field $\vec{E}$ given by :

$$\vec{E} = 100\,\hat{i}\,\frac{N}{C} \qquad \text{for } x > 0$$

$$= -\,100\,\hat{i}\,\frac{N}{C} \qquad \text{for } x < 0$$

exists in a region. A right circular cylinder of length 10 cm and radius 2 cm, is placed in the region such that its axis coincides with x-axis and its two faces are at x = – 5 cm and x = 5 cm. Calculate :

(a) the net outward flux through the cylinder, and

(b) the net charge inside the cylinder. *3*

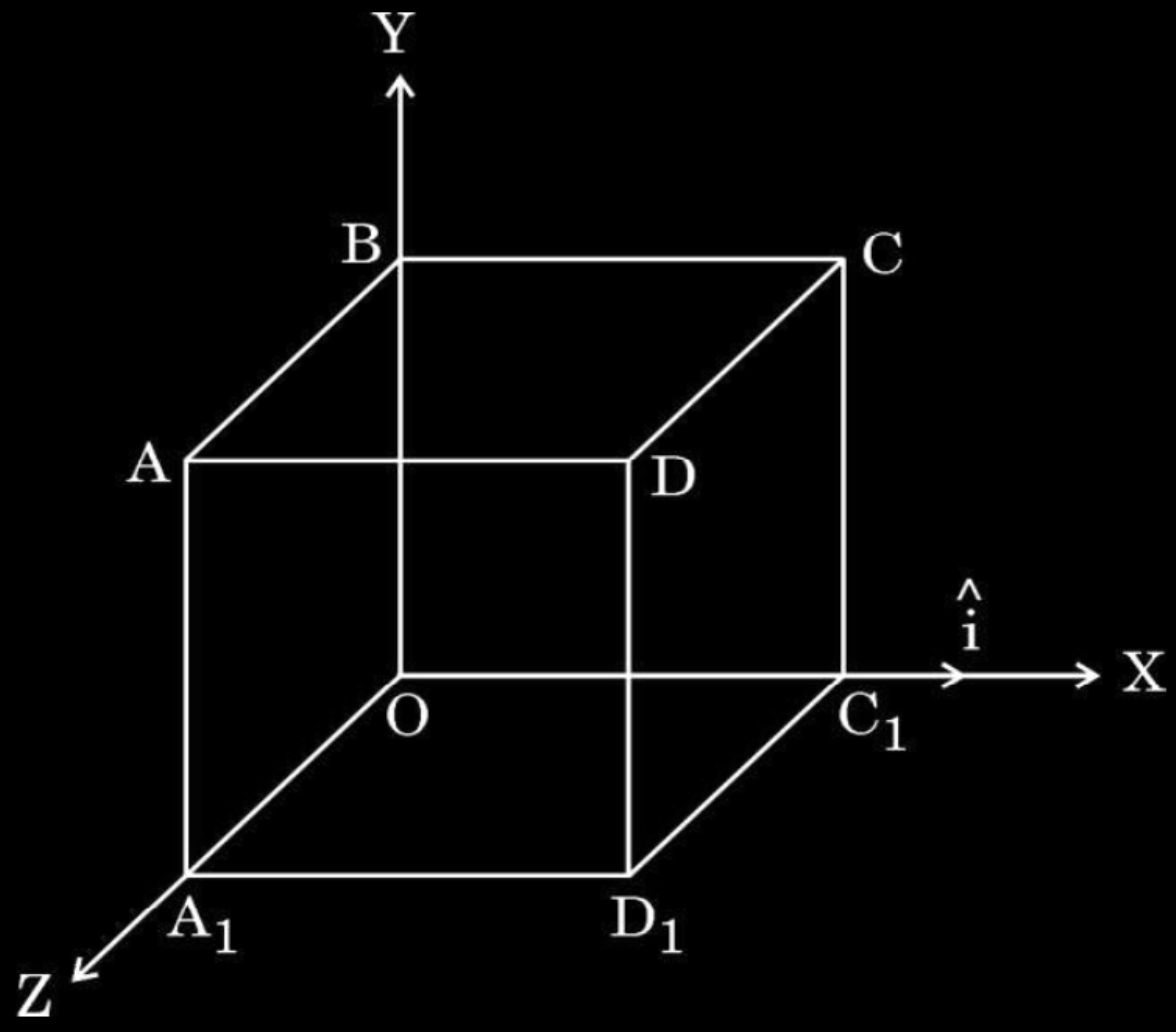
(b) (i) Obtain an expression for the electric field $\vec{E}$ due to a dipole of dipole moment $\vec{p}$ at a point on its equatorial plane and specify its direction. Hence, find the value of electric field :

(I) at the centre of the dipole ($r = 0$), and

(II) at a point $r >> a$,

where $2a$ is the length of the dipole.

(ii) An electric field $\vec{E} = (10x + 5)\hat{i}$ N/C exists in a region in which a cube of side L is kept as shown in the figure. Here x and L are in metres. Calculate the net flux through the cube.



3. A particle of mass m and charge q moving with velocity $v\hat{i}$ is subjected to a uniform electric field $E\hat{j}$. The particle will initially have a tendency to move in a circle of radius :

(A) $\left(\frac{mv}{qE}\right)$ in x-y plane

(B) $\left(\frac{mv^2}{qE^2}\right)$ in x-z plane

(C) $\left(\frac{mv^2}{qE}\right)$ in x-y plane

(D) $\left(\frac{mv}{qE^2}\right)$ in y-z plane

(B) The coulomb force between two point charges is 1.5×10^{-4} N. How will this force be affected in the following situations :

(i) the distance between the charges is doubled.

(ii) the magnitude of each charge is doubled but the distance between the charges remains the same.

1. A charge of $-1\ \mu C$ on a body represents **1**

(A) loss of 6.25×10^{12} electrons by the body.

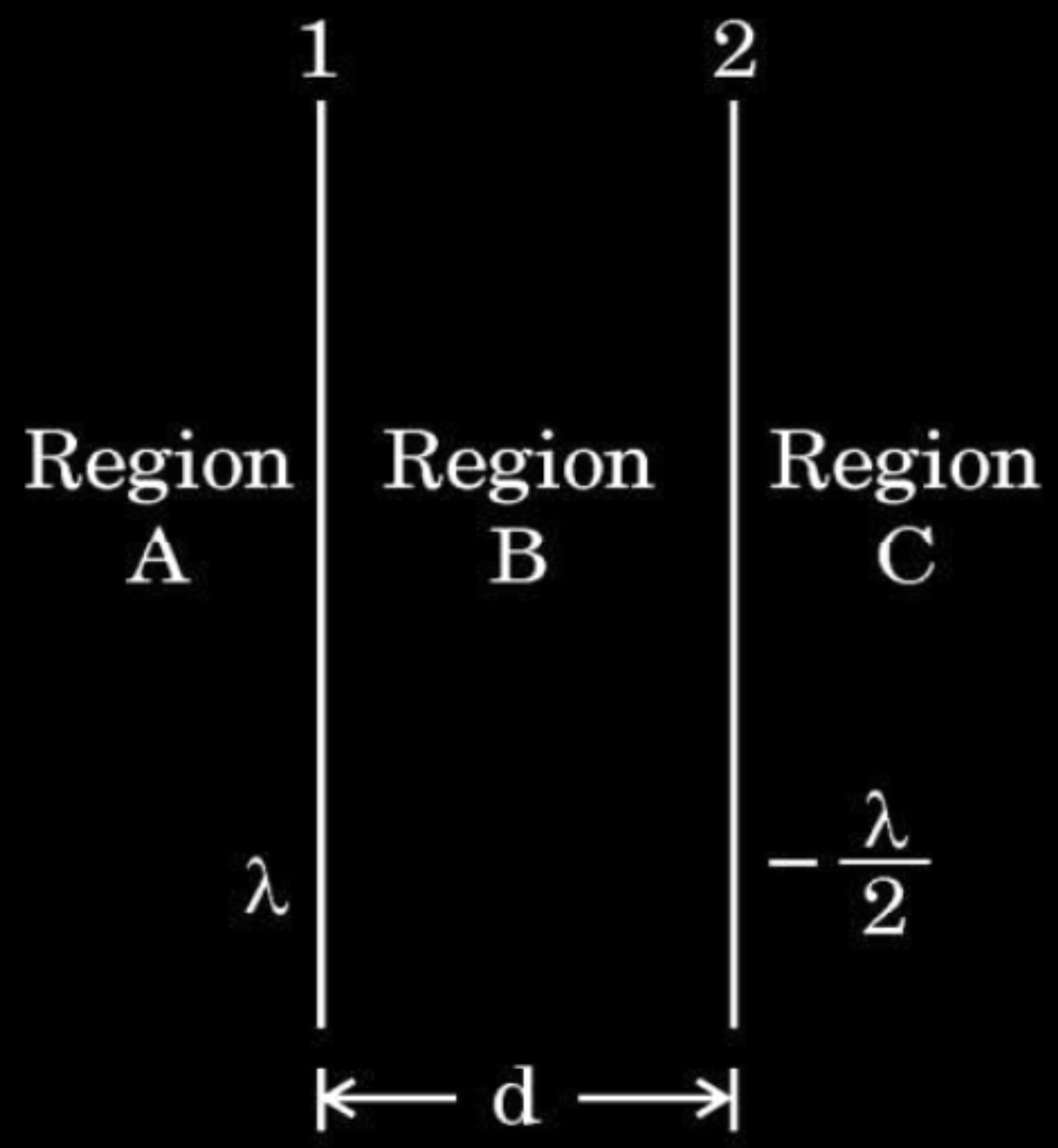
(B) gain of 6.25×10^{12} electrons by the body.

(C) gain of 1.6×10^{13} electrons by the body.

(D) loss of 1.6×10^{13} electrons by the body.

(ii) Two point charges q_1 (= 16 μC) and q_2(= 1 μC) are placed at points $\vec{r_1} = (3\ \text{m})\hat{i}$ and $\vec{r_2} = (4\ \text{m})\hat{j}$. Find the net electric field $\vec{E}$ at point $\vec{r} = (3\ \text{m})\hat{i} + (4\ \text{m})\hat{j}$.

(b) Two infinitely long straight wires '1' and '2' are placed d distance apart, parallel to each other, as shown in the figure. They are uniformly charged having charge densities λ and $-\frac{\lambda}{2}$ respectively. Locate the position of the point from wire '1' at which the net electric field is zero and identify the region in which it lies. *3*



(b) (i) Show that Gauss's theorem is consistent with Coulomb's law. Using it, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius r at a point at a distance y from the centre of the shell such that (I) $y > r$, and (II) $y < r$.

32. (a) (i) Two point charges $+ q$ and $- q$ are held at $(a, 0)$ and $(- a, 0)$ in x-y plane. Obtain an expression for the net electric field due to the charges at a point $(0, y)$. Hence, find electric field at a far off point ($y >> a$).

1. Two point charges Q and – q are held 'r' distance apart in free space. A uniform electric field $\vec{E}$ is applied in the region perpendicular to the line joining the two charges. Which one of the following angles will the direction of the net force acting on charge – q make with the line joining Q and – q ?

(A) $\tan^{-1} \dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(B) $\cot^{-1} \dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(C) $\tan^{-1} \dfrac{QE}{4\pi\varepsilon_0 r^2}$

(D) $\cot^{-1} \dfrac{QE}{4\pi\varepsilon_0 r^2}$

1. An electric dipole of dipole moment $1 \cdot 0 \times 10^{-12}$ Cm lies along x-axis. An electric field of magnitude $2 \cdot 0 \times 10^{4}$ NC^{-1} is switched on at an instant in the region. The unit vector along the electric field is $\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}$. The magnitude of the torque acting on the dipole at that instant is :

(A) $0 \cdot 5 \times 10^{-6}$ Nm

(B) $1 \cdot 0 \times 10^{-8}$ Nm

(C) $2 \cdot 0 \times 10^{-8}$ Nm

(D) $4 \cdot 0 \times 10^{-8}$ Nm

1. A body acquires charge $8{\cdot}0 \times 10^{-12}$ C. The mass of the body :

(A) increases by $4{\cdot}5 \times 10^{-7}$ kg

(B) decreases by $1{\cdot}0 \times 10^{-6}$ kg

(C) decreases by $4{\cdot}55 \times 10^{-23}$ kg

(D) increases by $9{\cdot}1 \times 10^{-23}$ kg

(b) (i) What is difference between an open surface and a closed surface ? Draw elementary surface vector $d\vec{S}$ for a spherical surface S.

(ii) Define electric flux through a surface. Give the significance of a Gaussian surface. A charge outside a Gaussian surface does not contribute to total electric flux through the surface. Why ?

(iii) A small spherical shell S_1 has point charges $q_1 = -3\ \mu C$, $q_2 = -2\ \mu C$ and $q_3 = 9\ \mu C$ inside it. This shell is enclosed by another big spherical shell S_2. A point charge Q is placed in between the two surfaces S_1 and S_2. If the electric flux through the surface S_2 is four times the flux through surface S_1, find charge Q.

1. Consider two identical dipoles D_1 and D_2. Charges $-q$ and q of dipole D_1 are located at (0, 0) and (a, 0) and that of dipole D_2 at (0, a) and (0, 2a) in x-y plane, respectively. The net dipole moment of the system is **1**

(A) $qa(\hat{i} + \hat{j})$

(B) $-qa(\hat{i} + \hat{j})$

(C) $qa(\hat{i} - \hat{j})$

(D) $-qa(\hat{i} - \hat{j})$

1. Two identical point charges are placed at the two vertices A and B of an equilateral triangle of side l. The magnitude of the electric field at the third vertex P is E. If a hollow conducting sphere of radius ($l/4$) is placed at P, the magnitude of the electric field at point P now becomes **1**

(A) $>E$

(B) E

(C) $\frac{E}{2}$

(D) zero

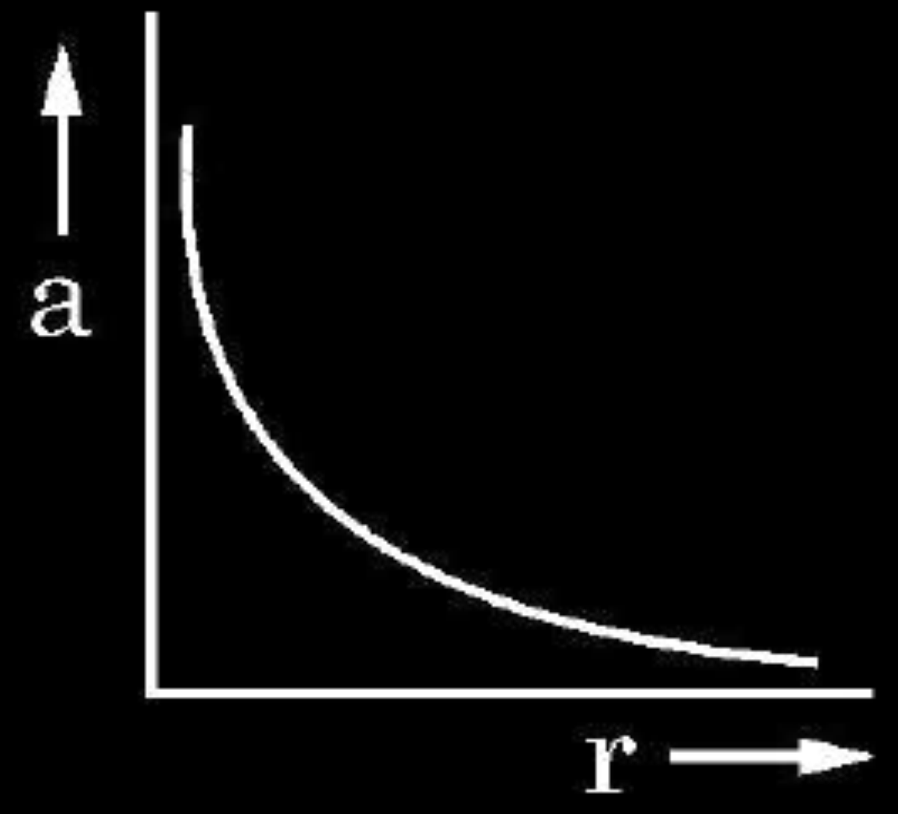
1. Two charges –q each are placed at the vertices A and B of an equilateral triangle ABC. If M is the mid-point of AB, the net electric field at C will point along **1**

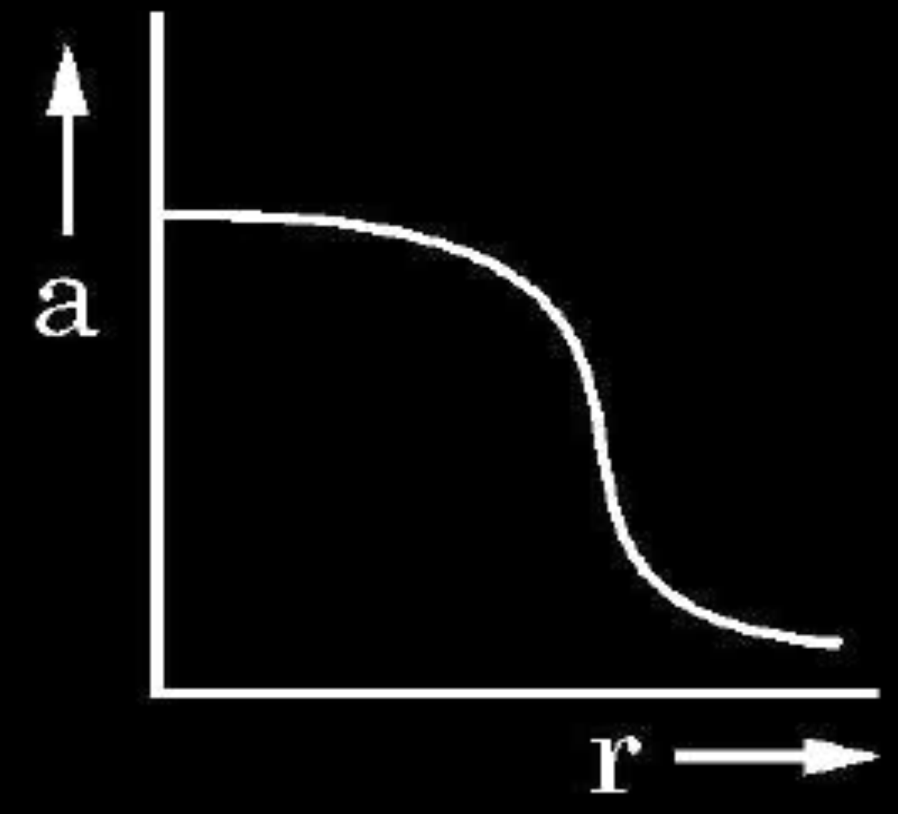
(A) CA (B) CB

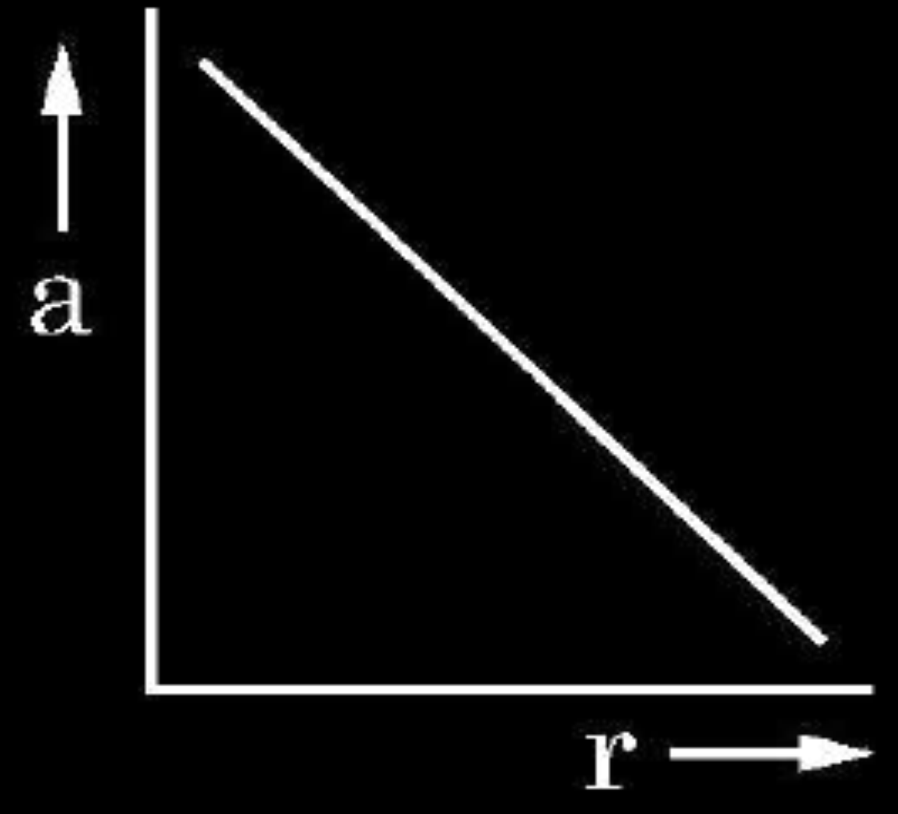
(C) MC (D) CM

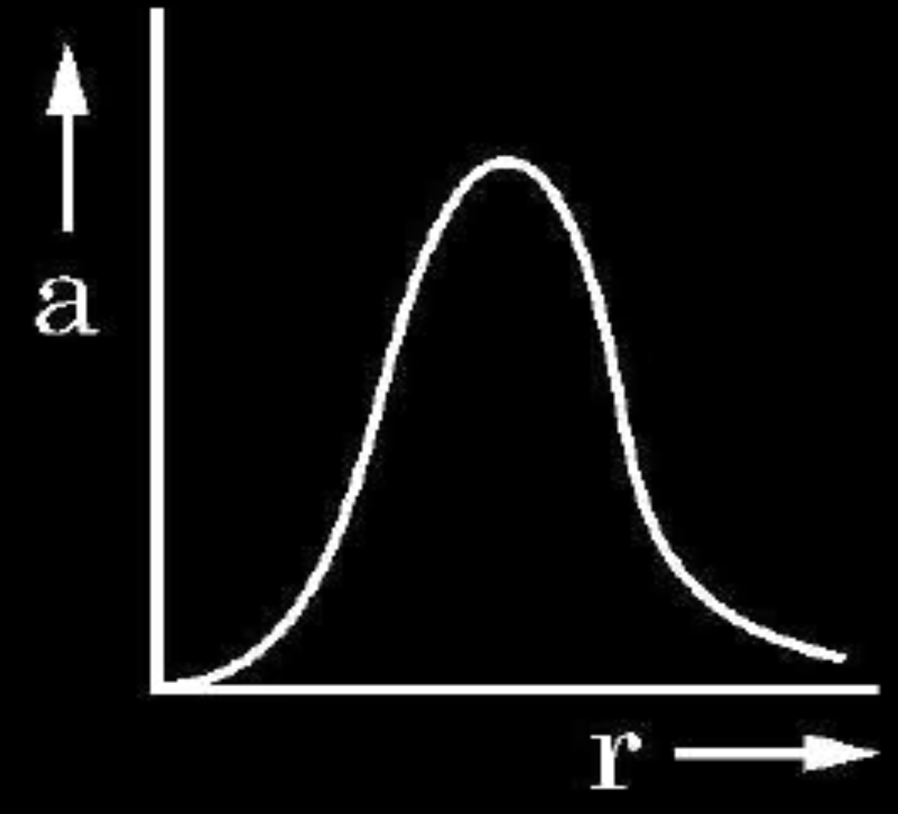
(b) (i) Consider three metal spherical shells A, B and C, each of radius R. Each shell is having a concentric metal ball of radius R/10. The spherical shells A, B and C are given charges +6q, –4q, and 14q respectively. Their inner metal balls are also given charges –2q, +8q and –10q respectively. Compare the magnitude of the electric fields due to shells A, B and C at a distance 3R from their centres.

1. A charge Q is fixed in position. Another charge q is brought near charge Q and released from rest. Which of the following graphs is the correct representation of the acceleration of the charge q as a function of its distance r from charge Q ? **1**

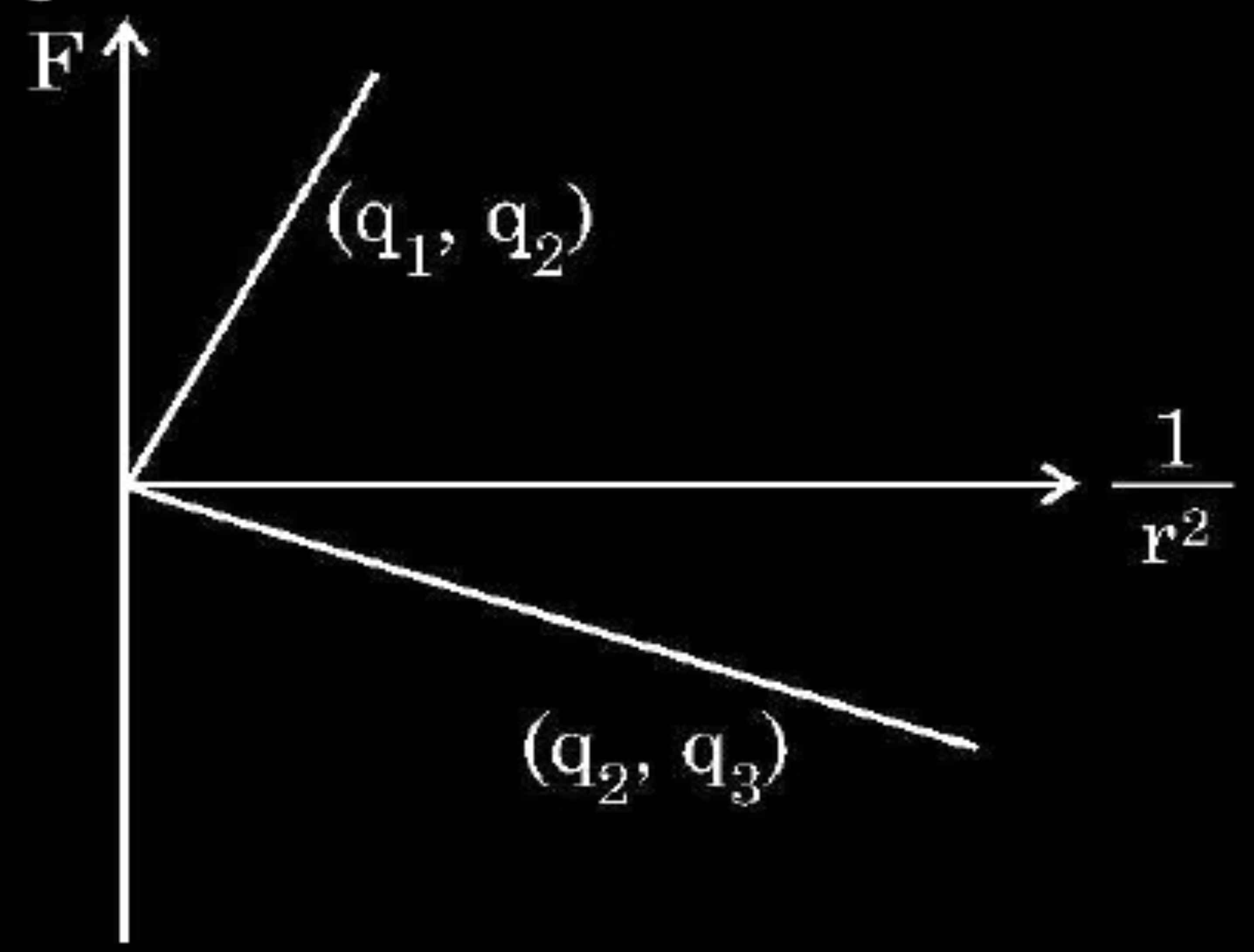
(A) 

(B) 

(C) 

(D) 

1. Figure shows variation of Coulomb force (F) acting between two point charges with $\frac{1}{r^2}$, r being the separation between the two charges (q_1, q_2) and (q_2, q_3). If q_2 is positive and least in magnitude, then the magnitudes of q_1, q_2 and q_3 are such that **1**



(A) $q_2 < q_3 < q_1$

(B) $q_3 < q_1 < q_2$

(C) $q_1 < q_2 < q_3$

(D) $q_2 < q_1 < q_3$